Paleomagnetism and orientation of Precambrian dykes, eastern Lake Superior region, and their use in estimates of crustal tilting

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Archean rocks form the eastern margin of the 1.1 Ga old Central North American rift along the eastern shore of Lake Superior and have been tilted westwards in response to rifting. Paleomagnetic and structural data from 2.6 Ga old Matachewan dykes suggest a westward crustal tilt of about 60°, which agrees well with dips recorded in nearby Keweenawan volcanics that rest directly on basement rocks. The Matachewan dyke swarm occurs throughout the east shore region of Lake Superior, whereas Keweenawan supracrustal sequences, which give a more precise estimate of tilt, are restricted to a few isolated shoreline patches. Estimates of crustal tilt can be obtained from the dykes on a regional basis, thus generating a more complete picture of basement deformation adjacent to a major intracratonic rift.

Les roches archéennes localisées sur le côté oriental du rift central de l'Amérique du Nord créé il y a 1,1 Ga, le long de la rive orientale du lac Supérieur, ont été dérangées et inclinées vers l'ouest par la formation du rift. Les données paléomagnétiques et structurales des dykes de Matachewan, âgés de 2,6 Ga, indiquent un mouvement d'inclinaison de la croûte vers l'ouest d'environ 60°, conformément aux pendages observés près des volcanites du Keweenawan qui reposent directement sur les roches du socle. L'essaim de dykes de Matachewan s'étend au travers toute la région de la rive orientale du lac Supérieur, tandis que les séquences supracrustales du Keweenawan, lesquelles fournissent des données plus précises sur l'inclinaison, n'apparaissent cependant que sous forme de quelques lambeaux isolés le long de la rive. Les estimations de l'inclinaison crustale peuvent être obtenues à partir des dykes considérés l'échelle régionale, fournissant ainsi une représentation plus complète de la déformation du socle adjacent à un rift intracratonique majeur.

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Introduction

The Central North American rift system (Ocola and Meyer 1973) is a major tectonic element of the southern Canadian Shield. It is mainly defined by basaltic lava flows and younger sediments that were deposited about 1.1 Ga ago in a subaerial to shallow-water environment along a zone of subsidence more than 2500 km long and up to 150 km wide. The volcanic and sedimentary rocks, known collectively as the Keweenawan Supergroup, outcrop only in the Lake Superior region, but they extend southwards beneath Phanerozoic rocks for many hundreds of kilometres from both ends of Lake Superior. Keweenawan rocks are interpreted as the product of an abortive rift, the volcanics representing the active crustal spreading phase and the sediments the subsequent infilling of the zone of subsidence as spreading ceased (e.g., Halls 1978). Towards the terminal stages of subsidence, compressional forces, perhaps related to Grenville tectonism, further accentuated the structure but also raised it locally along several major reverse faults existing along the limbs of the basin (Fig. 1).

The topic addressed in this paper concerns the extent to which the deformation of the Keweenawan rocks has involved the surrounding pre-Keweenawan basement. There appear to be two styles of deformation. The first is manifested by a progressive tilting of the basement towards the rift; the second, by a major fault that separates deformed rocks within the rift from those outside, which are relatively undisturbed. An example of the first case is found in the Thunder Bay area (Fig. 1). Here, flat-lying sedimentary rocks of the pre-Keweenawan Rove and Sibley sequences rest directly on Archean basement and are intruded by vertical Keweenawan dykes, which were presumably feeders for nearby flows. From the north shore of Nipigon Bay, where Sibley rocks are flat lying, there is a progressive increase in the dip of the Sibley and overlying Keweenawan Osler volcanics over a distance of about 30 km towards the south, reaching a maximum of more than 35° in the Osler volcanics at the mouth of Nipigon Strait (Halls 1974). An example of the second case can be found in the Ironwood area of Michigan and Wisconsin (Fig. 1). Here, Keweenawan Powder Mill volcanics dip steeply at about 80°N and rest almost conformably on Middle Precambrian, 2 Ga old quartzite and slate units that in turn unconformably overlie the Archean basement.[2] Numerous east-trending Keweenawan dykes with gentle southerly dips cut the Middle Precambrian units (Van Hise and Leith 1911, p. 236); these dykes were vertical before the host rocks were tilted. About 2 km south of the base of the Middle Precambrian sequence, horizontal Archean greenstones become vertical over a distance of about 100 m (M. Mudrey, personal communication, 1987). These relations suggest that about 6 km south of the base of the Keweenawan sequence there is a major fault separating rocks that are steeply tilted toward the rift axis from those to the south, which appear to be little affected by post-Keweenawan deformation. This fault is not found east of Lake Gogebic, where it may terminate against a northwest-trending cross fault (Fig. 1). East of Lake Gogebic, the basal Keweenawan volcanics have gentle dips of 20° to almost horizontal; in the Huron Mountains (Fig. 1), Keweenawan dykes of the Baraga swarm are close to vertical, suggesting little basement tilt in post-Keweenawan times. However, there is evidence that the crust may be gently tilted in this region because a crustal upwarp of at least 3 km occurs about 100 km farther south over a broad elliptical

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[2]Middle Precambrian is defined as the time interval between 2.6 and 1.6 Ga (van Eysinga 1975).

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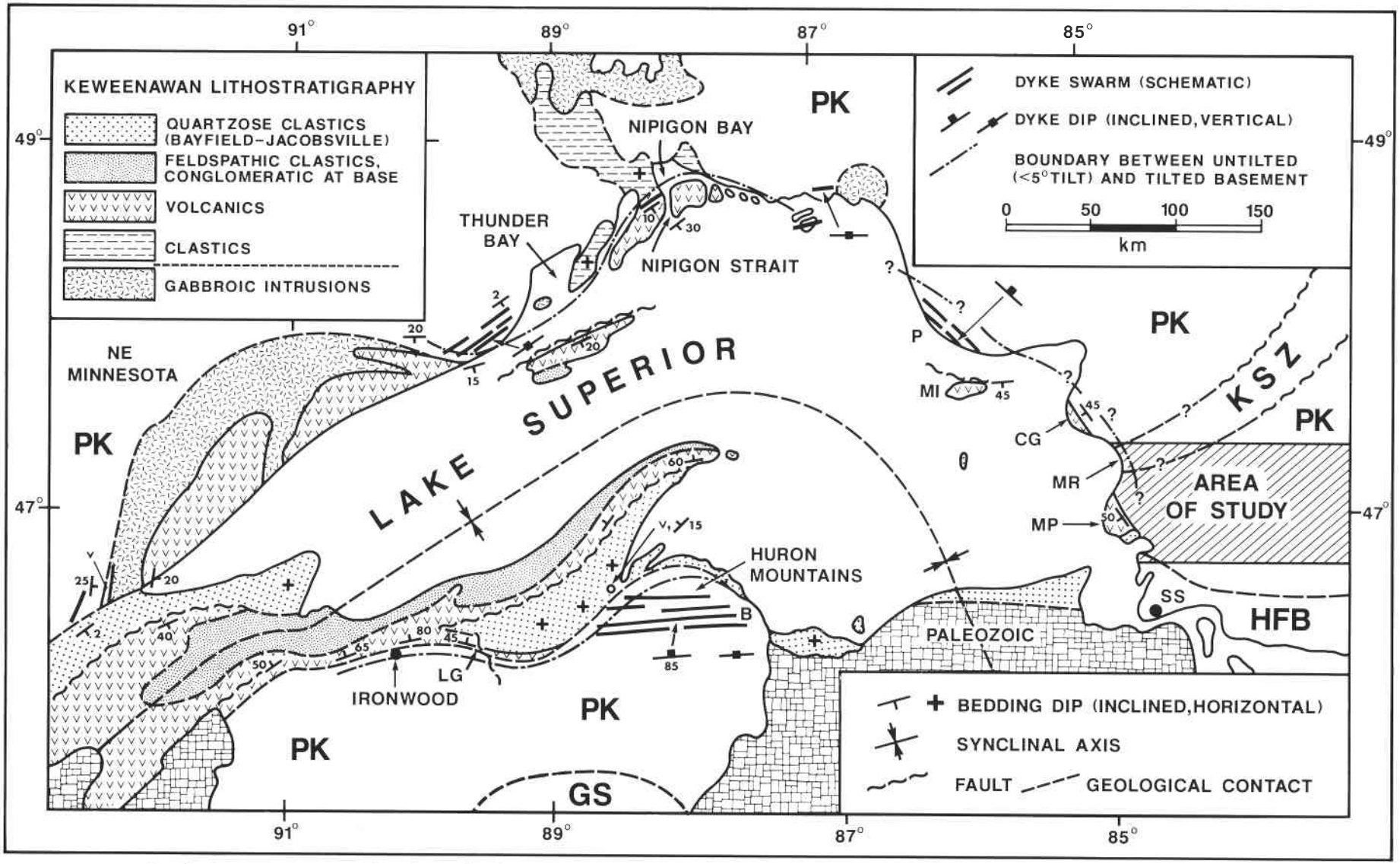


FIG. 1. General geological map of the Lake Superior region showing the form and position of the Central North American rift system and the location of the study area. Symbols: B, Baraga dyke swarm; LG, Lake Gogebic; P, Pukaskwa dyke swarm; MI, Michipicoten Island; CG, Cape Gargantua; MR, Montreal River Harbour; MP, Mamainse Point; SS, Sault Ste. Marie; PK, pre-Keweenawan basement; KSZ, Kapuskasing Structural Zone; HFB, Huronian fold belt; and GS, Goodman Swell (Peterman and Sims 1986).

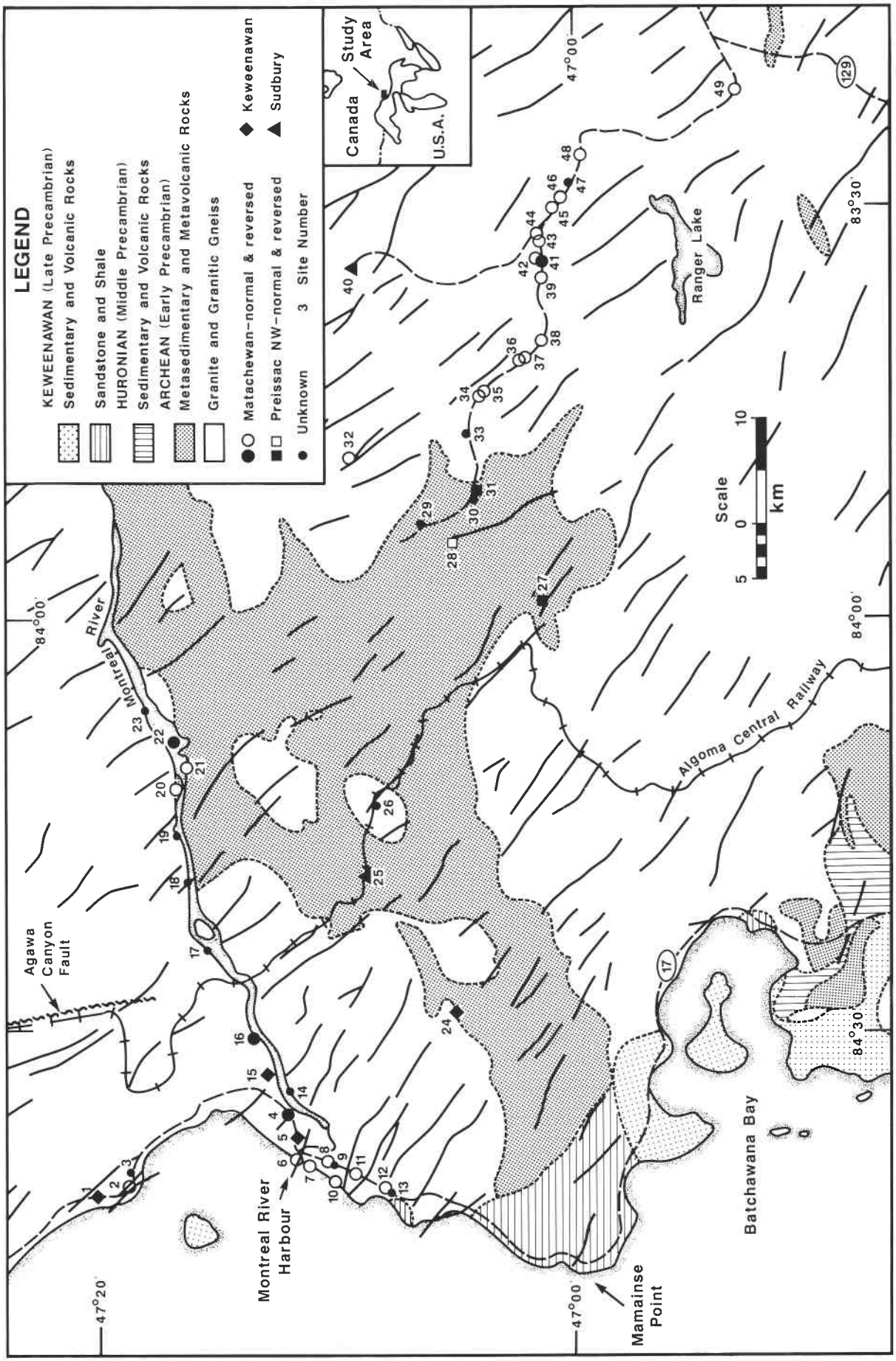


FIG. 2. General geological map showing paleomagnetic sampling sites coded according to swarm type based on lithology and paleomagnetism. Linear aeromagnetic anomalies caused by dykes (solid lines) give a north-northwest- to northwest-trending pattern. Note the slight fanning of dyke trends towards the northwest. Anomaly trends are more westerly close to the Lake Superior shoreline, where westward tilting of the crust has occurred.

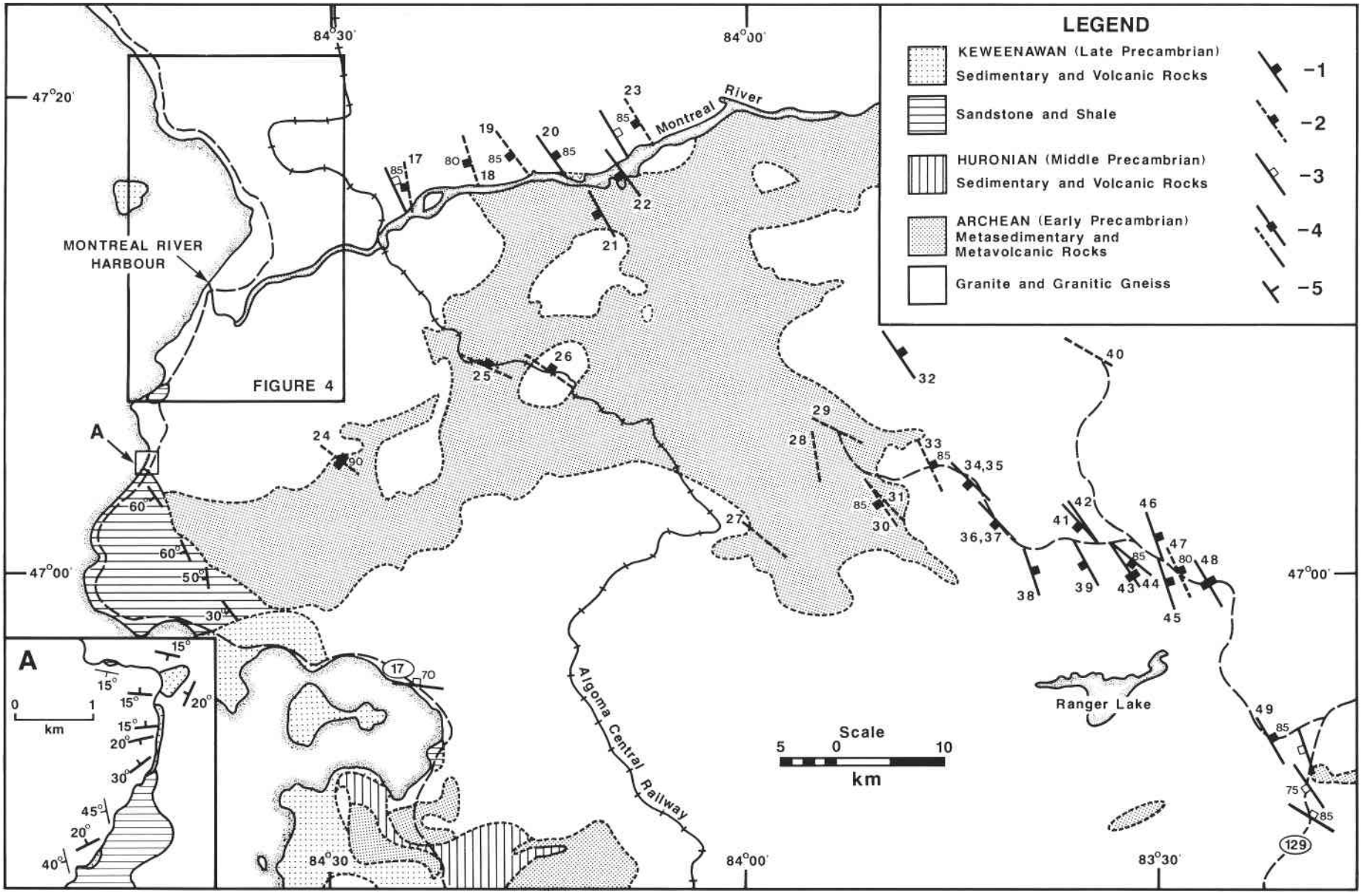


FIG. 3. Map of the study area showing structural data on dykes and Keweenawan supracrustals. Strike and dip symbols: 1, sampled Matachewan dykes; 2, sampled non-Matachewan dykes or dykes of unknown affiliation; 3, unsampled dykes; 4, dyke vertical, dip unknown; and 5, Keweenawan supracrustals. Large numbers beside symbols are site labels; small numbers are dip values, which are unspecified for dykes dipping less than 5° from vertical. The region of more abundant data around Montreal River Harbour is shown in Fig. 4. Inset map shows unconformable relations between middle and upper Keweenawan units in Mica Bay; thick (thin) dip symbols refer to Keweenawan clastic (volcanic) units.

region (GS in Fig. 1) of about 130 km × 60 km. This feature has been discovered on the basis of updated radiometric ages and has been termed the Goodman Swell by Peterman and Sims (1986). It is interpreted as a flanking forebulge of the lithospheric downwarp along the Mid-Continent rift, locally enhanced by the curvature of the rift axis (Peterman and Sims 1986).

In the eastern half of Lake Superior, the extent of shield deformation can be inferred locally from isolated patches of Keweenawan volcanics and sediments, such as those at Cape Gargantua and Mamainse Point (Fig. 1), which exhibit lakeward dips of more than 50° in their lower stratigraphic levels. However, in intervening regions devoid of Keweenawan supracrustals, knowledge of shield deformation must be based on more indirect evidence. For example, the northwest-trending Pukaskwa dykes (Fig. 1) dip at high angles to the northeast, and if they are restored to the vertical their paleomagnetic pole moves towards the Keweenawan polar wander path (Halls and Pesonen 1982).

In this paper, we study an area of shield in the vicinity of the Montreal River (Fig. 1) where Keweenawan supracrustals are virtually absent, in order to examine the feasibility of using paleomagnetism and the orientation of pre-Keweenawan dykes to provide an estimate of crustal tilting associated with the development of the Keweenawan rift.

General geology of the Montreal River area

The terrane south of the Montreal River (Fig. 2) is composed of Archean felsic plutons, gneisses, and complexly deformed supracrustal assemblages (Grunsky 1981, 1982). North of the Montreal River, migmatitic rocks may represent a southwesterly continuation of the Kapuskasing Structural Zone (KSZ in Fig. 1), and the river itself may follow a major fault separating amphibolite-facies rocks of the KSZ from terrane of lower metamorphic grade to the south (Grunsky 1981).

Keweenawan volcanic and sedimentary rocks outcrop locally along the Lake Superior shoreline and exhibit lakeward dips of 15–60° (Nuffield 1955; Annells 1973, 1974). At Mamainse Point, more than 5000 m of volcanics and interflow sediments is exposed. Younger Keweenawan sedimentary rocks may be found on offshore islands and locally along the mainland shore. At both Mica and Alona bays (inset map, Figs. 3 and 4), they dip lakeward and rest unconformably on Keweenawan volcanics, but elsewhere, on offshore islands and in the vicinity of Sault Ste. Marie, beds are gently dipping to flat lying.

Detailed mapping along the shoreline south of the Montreal River by Nuffield (1955) revealed many northwest-trending diabase dykes with northeasterly dips, inviting speculation that the crust here acquired a westerly tilt during Keweenawan basin subsidence. Aeromagnetic maps show a profusion of dykes (Fig. 2) crossing the Archean shield east of Lake Superior. Nearly all have northwesterly trends, indicating that the dyke set mapped by Nuffield is of great regional extent.

Dyke swarms are widespread in the southern Canadian Shield, and some of them have spectacular dimensions. The most extensive one, the 2.6 Ga old Matachewan swarm, trends north to northwest and is composed of quartz tholeiite dykes that are found in virtually every part of a 500 km wide swath of Archean shield that borders Lake Superior to the east and northeast. These dykes have been documented, mainly through paleomagnetic studies, in the Abitibi Belt (e.g., Strangway 1964; Pesonen 1973; Buchan and Schwarz 1981; Condie *et al.* 1987) and also in a region extending from Hearst to Wawa on the western side of the KSZ (Ernst and Halls 1984). Matachewan dykes are recognized in the field by the presence of pale green feldspar phenocrysts and in the laboratory by a dual magnetic polarity (declination: $D = 30°/210°$; inclination: $I = +10°/-10°$). Their paleomagnetic signature is unique amongst Canadian Shield dykes and serves to distinguish them, especially where they lack phenocrysts, from dykes of younger swarms. Although many of the dykes shown in Fig. 2 are likely to be Matachewan, others of similar trend, but belonging to three younger groups, may be present. North of the Huronian fold belt (HFB in Fig. 1) in the Elliot Lake area, Aibangbee (1982) mapped northwest-trending quartz diabase dykes that both predate and postdate the basal Huronian sedimentary units. The pre-Huronian dykes give typical Matachewan paleomagnetic signatures, whereas the post-Huronian dykes (provisionally referred to as "Preissac NW") resemble, lithologically and paleomagnetically, the 2.1 Ga old northeast-trending Preissac quartz tholeiites. Alkali olivine basalt dykes belonging to the 1.25 Ga Sudbury swarm extend in a west-northwest direction across the Sudbury region and may continue into the area shown in Fig. 2. These dykes can be readily distinguished not only by their freshness and by the presence of olivine, but also by their characteristic paleomagnetic direction ($D = 270°$; $I = 0°$; e.g., Fahrig *et al.* 1965; Palmer *et al.* 1977). Olivine tholeiite dykes associated with Keweenawan igneous activity have been reported along the east shore of Lake Superior (e.g., Ayres 1969; Giblin 1973). They have a general north-northwest trend, in keeping with the overall north-northwest trend of the adjacent rift (Halls 1978), but their abundance relative to dykes of other ages is unknown.

Field procedure

Forty-nine paleomagnetic sites were obtained from northwest-trending dykes along two traverses approximately normal to the Lake Superior shoreline. At least six block samples or field-drilled cores were collected at each site and oriented using both a sun compass and a magnetic compass. Particular attention was paid to collecting samples representative of both chilled margins and interiors, and differences between sun and magnetic compass readings were used to help avoid lightning-struck outcrops. The locations of paleomagnetic sites (Fig. 2) provide a profile extending from areas of suspected Keweenawan deformation close to Lake Superior to regions more than 100 km inland. Therefore, the nature and extent of shield deformation can be studied as the Keweenawan rift is approached from the east. A by-product of this study is an assessment of the relative frequency of dykes belonging to the various age groups. The thickness and dip of the dykes were measured where possible at each paleomagnetic site and at several shoreline locations where samples were not collected (Appendix). The structural information is shown in Figs. 3 and 4.

Experimental procedure

From each paleomagnetic sample, at least one cylindrical core (2.54 cm in length and diameter) was subjected to stepwise alternating-field demagnetization using a Schonstedt single-axis demagnetizer, followed by measurement of the remanence direction using a modified DIGICO magnetometer.

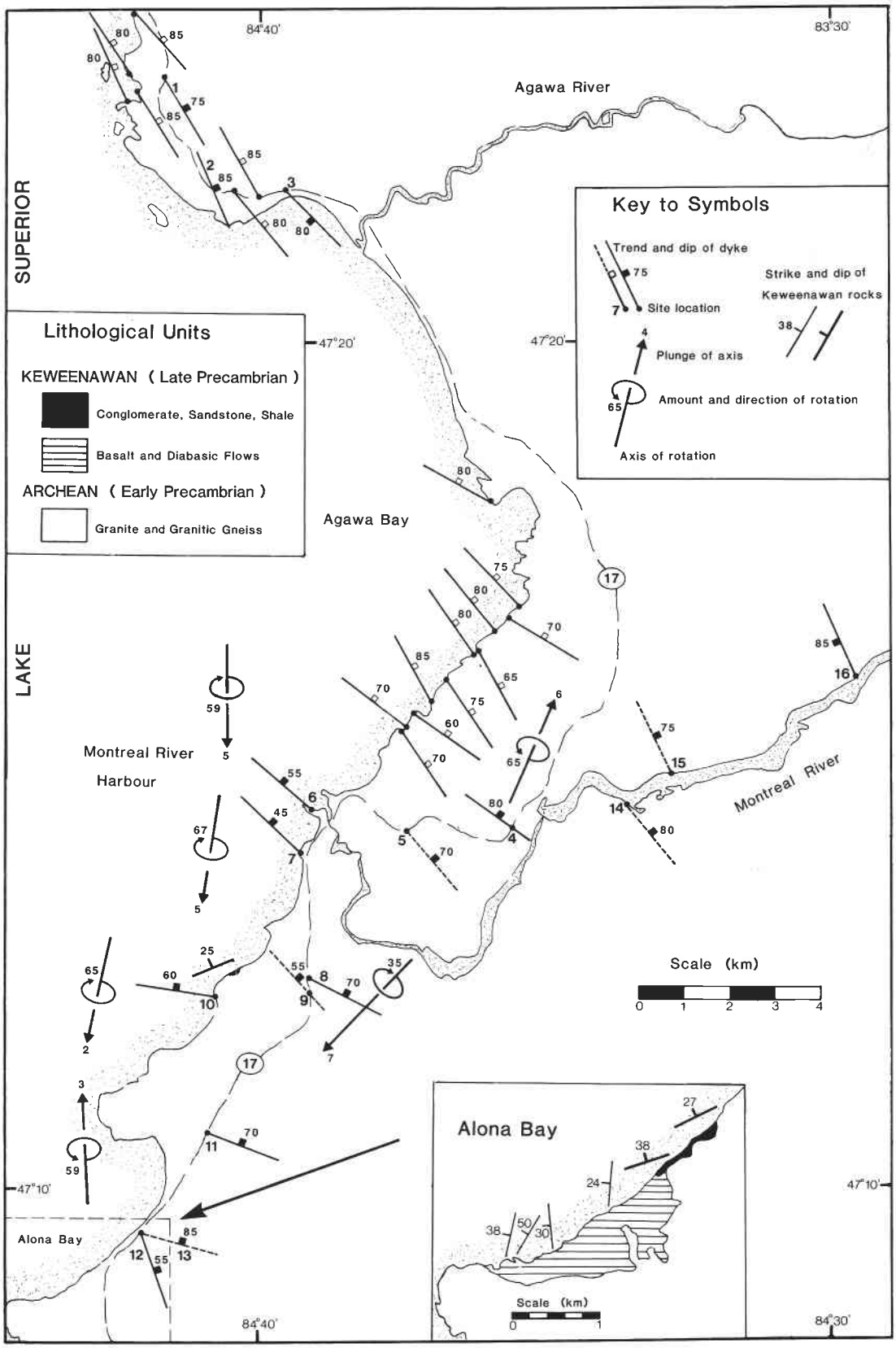


FIG. 4. Map of Montreal River Harbour area showing structural data on dykes and Keweenawan supracrustals. Strike and dip symbols as in Fig. 3. Inset map shows unconformable relations between middle and upper Keweenawan units in Alona Bay. Note small coastal patch of upper Keweenawan clastics dipping northwest about 4 km south of the Montreal River. Axes and amounts of crustal rotation, calculated using structural and paleomagnetic data from Matachewan dykes, are also shown.

From 8 to 20 demagnetization steps were performed on each specimen, depending on the complexity of the remanence signature. Demagnetization and magnetometer measurements were terminated for each specimen when the field limit of the demagnetizer (100 mT) was reached or when remanence directions and intensities began to scatter. Specimen demagnetization data were analyzed using methods described by Halls (1986).

Paleomagnetic results

On the basis of characteristic components of magnetization, lithology, and petrography, the dykes were divided into four main age groups corresponding to the Matachewan, Preissac NW, Sudbury, and Keweenawan swarms (Appendix). Within the Matachewan and Preissac NW swarms, dykes of both normal and reversed magnetic polarity were found. Details of specimen demagnetization behaviour and petrography were given by Shaw (1985). Matachewan dykes are by far the most numerous. Lithologically, they are quartz diabases, often containing the diagnostic green plagioclase phenocrysts. In general, they are moderately to heavily altered (50−80%), with groundmass plagioclase replaced by sausserite and chlorite, and pyroxene extensively uralitized, especially around the margins. Despite alteration, however, the lath shape and twining of plagioclase are still visible in most dykes. Two dykes, at sites 2 and 4, were remarkably fresh and exhibited higher natural remanent magnetization (NRM) intensities and more stable, higher coercivity characteristic components that gave well-defined site directions (Appendix). In the altered dykes, only about 10% of NRM intensity remained after 20 mT demagnetization, with further demagnetization required to yield a stable end point. Compared with the fresh dykes, however, stable end points were defined with lower precision (mean angular deviation was $\leq 10°$ compared with $\leq 5°$). Many altered dykes, lithologically indistinguishable from Matachewan types, particularly those along the Montreal River, did not contain any coherent magnetization (Appendix), whereas others yielded poorly determined directions based on only one or two samples, often from chilled margins.

Despite the foregoing problems, it is clear that the Matachewan population, apart from reversals, divides into two magnetic groups: one comprising dykes within about 3 km of the Lake Superior shoreline, and the other from more inland sites. Only the latter gave a mean direction typical of that obtained elsewhere for Matachewan dykes (Fig. 5*a*). These inland dykes all dip within about 5° of the vertical, whereas those peripheral to the lake have northeasterly dips between 40 and 80° (Fig. 5*b*; Appendix). The interpretation of this result, discussed later in more detail, is that coastal dykes have been tilted to the west relative to interior dykes. Also, two coastal dykes, sites 1 and 5, with pronounced northeasterly dips and lithologically similar to Keweenawan basalts, yielded paleomagnetic directions close to those expected for Keweenawan rocks ($D = 300°$; $I = +40°$; e.g., Halls and Pesonen 1982) but with inclinations about 20° steeper than expected, again in accord with their having been tilted to the west.

Regional variations in dyke deformation

For all inland dykes, there is little evidence of deformation subsequent to emplacement; occasionally, slip lineations are seen on joint surfaces, but they are usually faint and poorly developed. In contrast, the coastal dykes, especially those dipping less than 70°NE, exhibit more intense and pervasive slip lineations, the gouging along joint surfaces being deeper and wider. Some of the dykes have superimposed sets of lineations and are extensively sheared along dyke-parallel joints, especially those close to chilled margins. Narrow, centimetre-wide zones of more finely comminuted material are often seen where shearing has been more severe. Breccia zones are also found, but they are not necessarily dyke parallel, and veins of calcite and epidote are common, especially near dyke margins. Our preliminary study indicates, therefore, that coastal dykes of Matachewan affinity have suffered more shearing and brecciation than their more inland counterparts. However, detailed measurement of the orientation and sense of displacement on slip surfaces, which would place further constraints on the deformation history, has not been made.

Brecciation, gouging, and intense hematization along fractures have also been observed in granitic host rocks in the Alona Bay area, but these are not features of the Archean basement farther inland (E. Grunsky, personal communication, 1987). Hematization was also observed in several coastal dykes, being particularly severe at site 14.

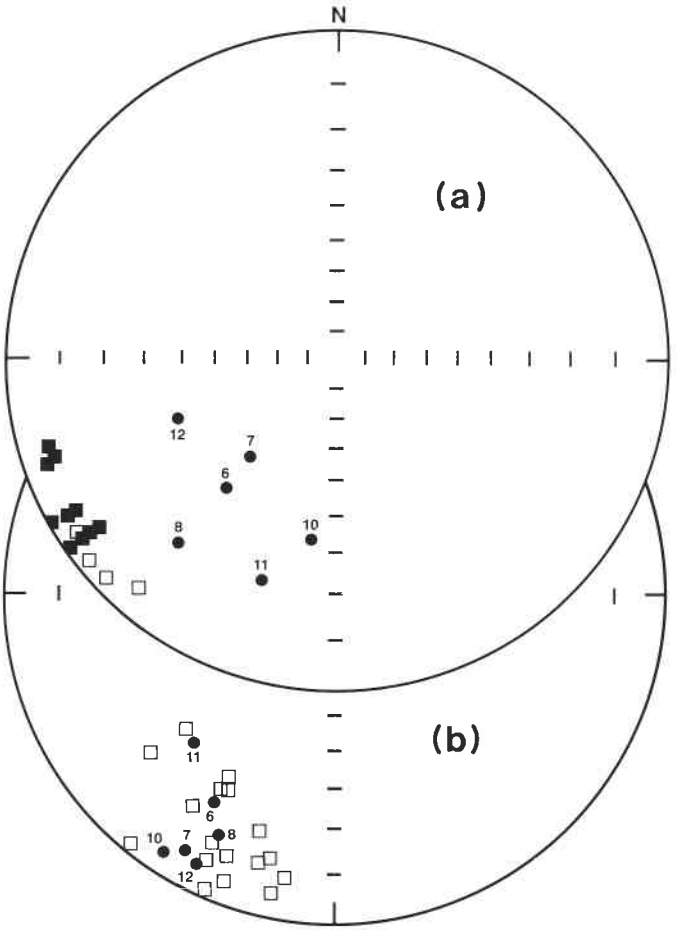


FIG. 5. (*a*) Poles to dyke planes and (*b*) paleomagnetic directions, plotted for reversely magnetized Matachewan dykes. In both diagrams, unrotated (rotated) dykes are represented by squares (circles). Solid (open) symbols are plotted on the lower (upper) hemisphere.

Calculation of crustal tilt

Dyke attitude and paleomagnetic direction were used as shown in Fig. 6 to provide an estimate of the degree and direc-

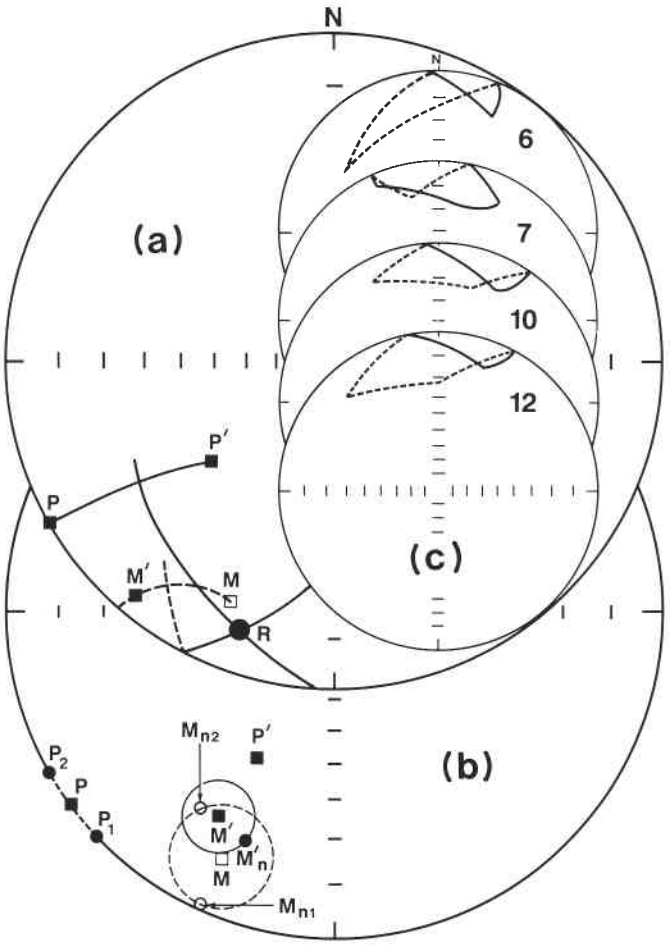


FIG. 6. (*a*) Construction for finding the axis and amount of rotation given rotated and unrotated pairs of points representing the pole to the dyke plane and paleomagnetic direction. The rotation axis is the intersection (R) of two planes that are the normal bisectors of the angles between the two poles (P, P') and the two magnetization directions (M, M'), where the primed values refer to the rotated state. The amount (and sense) of rotation about R is that required to bring the two poles and the two magnetization directions into simultaneous coincidence. (*b*) Example of the construction used to determine the envelope of possible rotation axes. P and M are mean values obtained from several unrotated dykes (Appendix). P' is a single orientation measurement of a rotated dyke at the outcrop site. M', the magnetization direction of the dyke, is a mean of several sample determinations. Associated with both M and M' are circles drawn around the mean values that have a radius of one (circular) standard deviation (θ_{63}). The inclination of P averages within a degree of the horizontal, with relatively little spread compared with the declination. A measure of the dispersion of P is thus given by P_1 and P_2, which are at one standard deviation of declination on either side of P. An arbitrary point M'_n on the M' circle is first selected at an angular distance β from P'. Using P_1 and the condition that β remains constant during rotation, only two possible values of M, namely M_{n1} and M_{n2}, are possible. These values are then used, together with P', P_1, and M'_n, to generate two poles of rotation through the construction in (*a*). The above procedure is then repeated for M'_n every 9° around the M' circle, and then the whole process is repeated using P_2. The resultant envelopes of rotation axes are shown for four reversed Matachewan dykes in (*c*), in which dashed (solid) lines are for the lower (upper) hemisphere. The axis orientation and rotation values given in Table 1 are means of all determinations defining each envelope. A computer program to calculate rotation axes using the above procedure was given by Shaw (1985).

TABLE 1. Rotation calculations for Matachewan dykes

Dyke	Axial trend (°)	Axial plunge (°)	Rotation (°W)
4	017 (016)	6N (2S)	65 (54)
6	356 (313)	5S (10S)	59 (69)
7	002 (350)	5S (11S)	67 (65)
8	045 (054)	7S (5N)	35 (58)
10	011 (356)	2S (8S)	65 (58)
12	356 (337)	3N (5S)	59 (52)
Mean	010 (358)	2S (5S)	58 (59)

NOTES: Reference (unrotated) dyke parameters (see Fig. 6 for key to symbols): $M = 204°/-11.6°$; $\theta_{63} = 11.2°$; P_1, $P_2 = 225°/0°$, 245°/0° (205°/0°, 225°/0°). Figures in parentheses are for calculations based on a more west-northwest trend for the inland dyke population. For values of M' see Appendix; θ_{63} values for rotated dykes are as follows: dyke 4, 8°; dyke 6, 17°; dyke 7, 16°; dyke 8, 6°; dyke 10, 7°; and dyke 12, 10°. P' values are derived from dyke trends and dips given in the Appendix, except that the mean trend value (307°) for rotated dykes was assigned to the dykes at sites 10 and 12, where anomalous trends, probably due to dyke branching, were observed.

tion of crustal tilting along the eastern margin of the Keweenawan rift.

The results obtained for six dykes are similar (Fig. 4; Table 1), but no valid rotations could be made for dyke 11 because the greatest angle possible between the rotated magnetic direction and the pole to the rotated dyke plane was smaller than the least angle possible between the unrotated magnetic direction and the pole to the unrotated dyke plane. A solution could be obtained by relaxing the requirement of equality of β (the angle between the remanence direction and the pole to the dyke plane) between rotated and unrotated dykes (a reasonable expectation because of secular variation), but we have not incorporated this additional degree of variability into the method of Fig. 6. Because we do not know the original strike and dip for any given dyke, we have not pursued the calculation further. However, the close agreement between solutions for six dykes suggests that the results are meaningful. They indicate rotation axes that are within 10° of horizontal with an average trend of about 000°−010° and suggest a rotation of ~60°W, which is close to the maximum value of dip seen in the basal sections of the Mamainse Point sequence (cf. Figs. 3 and 4). In Table 1, two sets of calculations are given. The first uses reference parameters that correspond to those measured for the inland dykes. For inland dykes that are on strike with those near Montreal River harbour, aeromagnetic data show (Fig. 2) that their trend is more west-northwest than that of the sampled inland dykes. The second calculation in Table 1 uses this different trend value but otherwise keeps the same reference parameters as before. A comparison of the rotation parameters obtained from the two data sets shows that they are relatively insensitive to a 20° change in the regional strike of the swarm.

The calculated rotation parameters represent a composite of all rotations that have affected the dykes. We know from the relationships at Mica and Alona bays that tilting to the southwest of middle Keweenawan volcanic sequences was followed by a more northwesterly tilting after upper Keweenawan rocks were formed, and it is noteworthy that most of the calculated

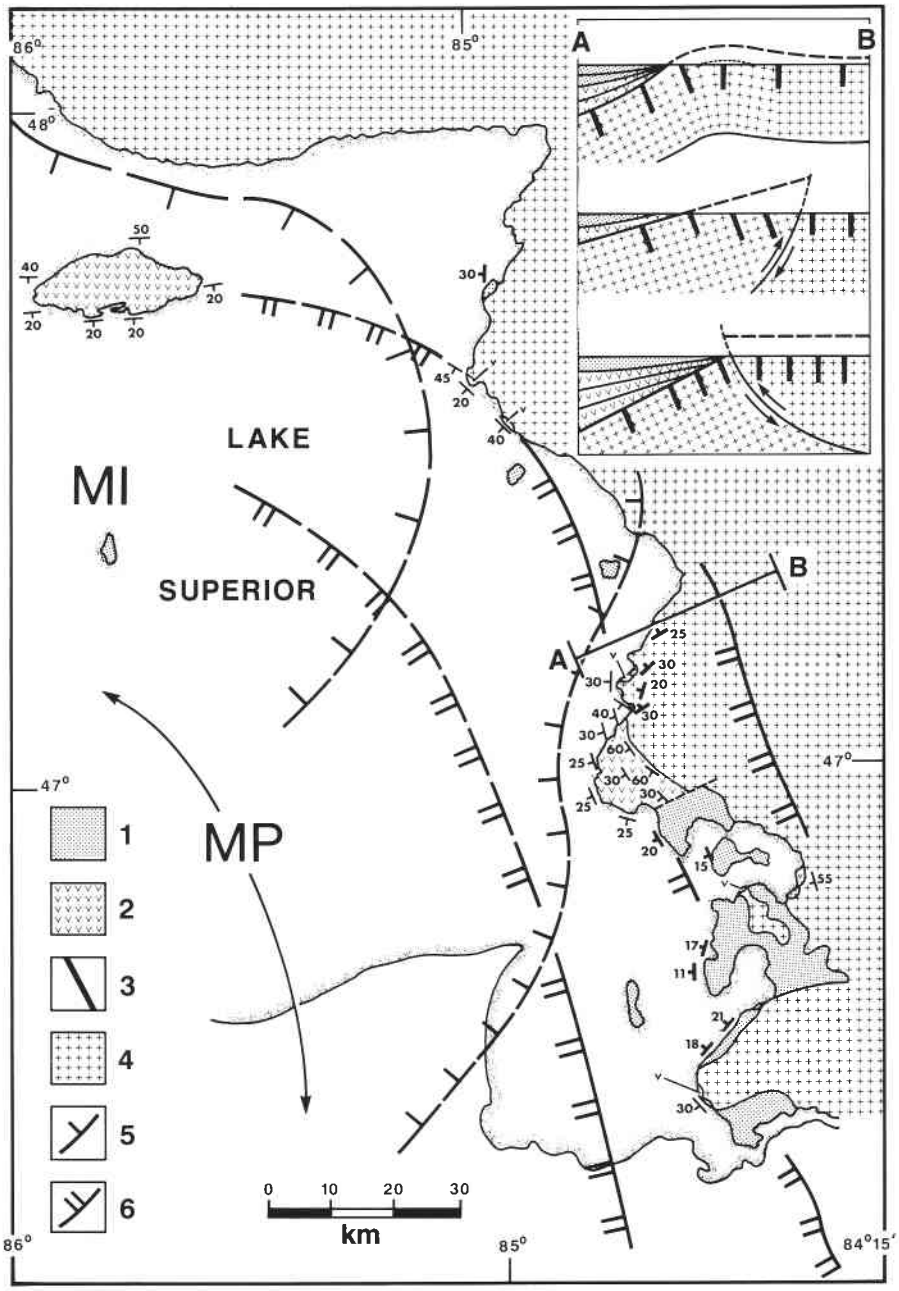


FIG. 7. Keweenawan volcanic plateaus of the eastern Lake Superior region, modified after White (1972) and Green (1977). Plateau names: MI, Michipicoten Island; MP, Mamainse Point. Legend: 1, upper Keweenawan clastics; 2, middle Keweenawan volcanics; 3, dykes in schematic sections; 4, pre-Keweenawan granitic basement; 5 and 6, structural form lines of basins centred on the Michipicoten Island and Mamainse Point plateaus. Thin (thick) dip symbols refer to Keweenawan volcanic (clastic) units. The MI sequence comprises flows that on the basis of paleomagnetism and geochemistry are mainly younger than the MP sequence (Annells 1974). Inset diagram shows, schematically, possible geological cross sections along profile A–B through the Montreal River harbour area.

rotation axes have trends that are intermediate between the observed strike directions in middle and upper Keweenawan units. The large value of tilt obtained may reflect a progressive lakeward tilting that occurred throughout Keweenawan time after initial volcanism. This progressive deformation is suggested by a fanning of dips to lower values at higher stratigraphic levels within the Mamainse Point sequence (Fig. 3).

The pronounced sawtooth outline of the Lake Superior east coast (Fig. 1) suggests a strong control in its formation by northwest- and northeast-oriented joints. The middle Keweenawan volcanic sequences at Cape Gargantua and Mamainse Point both dip to the southwest and are found along southwest-facing shorelines, whereas upper Keweenawan clastics that dip northwest are found along northwest-facing coasts (Figs. 3 and 4). The deformation and preservation of these supracrustal patches may thus have been determined in part by faulting along the northeast- and northwest-striking joint planes. The geological relations are consistent with there having been a change in the position of maximum subsidence foci in the eastern Lake Superior basin. Two volcanic plateaus (Fig. 7) have been advocated in this region (White 1972; Green 1977): an older, southerly one, of which the Cape Gargantua and

Mamainse Point sequences are part; and a mostly younger, more northerly one, now represented by the Michipicoten Island volcanic sequence (Fig. 7). It has been demonstrated elsewhere along the Keweenawan rift (see Halls 1978 for summary) that upper Keweenawan clastics are often thicker over areas of anomalously thick volcanics, a possible consequence of isostatic subsidence in areas of greatest volcanic activity. The structural relations observed along the east coast may thus reflect a change in the position of maximum volcanism (and hence crustal subsidence) from southeast to northeast Lake Superior between middle and late Keweenawan times.

The regionally extensive nature of the Matachewan dyke swarm allows an estimate to be made of the width of the surrounding shield that was tilted during Keweenawan rift development. The eastern boundary is best defined along the Montreal River and is marked by a transition from coastal dykes that dip consistently to the northeast to interior ones that dip within 5° of vertical to either the northeast or southwest (Fig. 4). The boundary lies somewhere between sites 15 and 16, thus suggesting that a rim of shield no more than 6–10 km has been rotated. The boundary lies approximately on a southerly extension of the north-trending Agawa Canyon fault (Fig. 2), of unknown nature and age. Further dyke data profiles are required at other locations along the east coast in order to define the position of the boundary on a more regional basis, and until then the exact mechanism by which tilting has been achieved remains conjectural. The dominant northwest and northeast strikes shown by Keweenawan supracrustal sequences suggest that tilting has involved movement along northwest- and northeast-trending faults. A listric reverse nature of these faults as shown in Fig. 7 (inset) would facilitate a westward crustal tilting. If this tilting is in response to a presumed axis of crustal loading, the generation of reverse faults might imply that crustal subsidence was accompanied by compression. Listric normal faults inclined towards the rift axis would lead to eastward dips rather than the observed westward dips. An alternative interpretation (inset, Fig. 7) is that the whole crust has been involved in flexuring, although the rapid increase in tilt from zero to 60° in less than 10 km would not favour this model. Further structural and paleomagnetic studies are required before these issues can be resolved.

Conclusions

Paleomagnetic and structural data from dyke swarms offer an important means to measure crustal tilting in areas devoid of supracrustal cover rocks formed prior to the tilting episode. Along the east coast of Lake Superior, we have demonstrated the power of this method through observations of 2.6 Ga old Matachewan dykes that are both areally extensive and older than regional deformation associated with the development of the Keweenawan rift system. The abundance of these dykes around the northern edge of the Huronian fold belt offers the possibility that adjacent shield deformation can likewise be established, particularly because basal Huronian units dip steeply over much of their outcrop length.

Our results demonstrate the potential of using dykes to estimate crustal deformation around the eastern margin of the Keweenawan rift. Detailed examination of only a small section of coastline north of the Montreal River reveals an abundance of northeast-dipping dykes that were not sampled (Fig. 4); thus, the statistics of our tilt determinations could be improved considerably by further structural and paleomagnetic data on a larger dyke population. We found that in the vicinity of Montreal River Harbour the crust has been tilted about 60° to the west, a result that is a composite of at least two tilting episodes that may be related to a change in the area of maximum subsidence within the Lake Superior basin. Only along the Montreal River can we define the position of the transition zone between tilted and untilted crust. Here, it is about 6–10 km from the shoreline. The position and sharpness of this boundary along the east shore of Lake Superior, if defined, would reveal much on a regional basis about the extent and nature of pre-existing faults in determining the direction and amount of crustal tilting.

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Appendix

TABLE A1. Dyke location, attitude, and paleomagnetic data

Site	Lat. (°)	Long. (°)	Trend (°)	Dip (°)	Width (m)	NT	NS	NR	D (°)	I (°)	D' (°)	I' (°)	Plat. (°)	Plong. (°)	α_{95} (°)	k
MATACHEWAN DYKES																
Reversed polarity, unrotated																
2	47.36	−84.58	325	85N	6	8	6	1	214.7	−14.8	214.3	−14.8	−40.65	−132.07	4.4	207
20	47.32	−84.20	325	85N	31	6	3	0	230.2	−16.9	230.1	−17.1	−32.81	−148.59	22.8	30
21	47.32	−84.18	330	>85N	11	7	4	0	209.7	−22.1	209.7	−22.5	−46.54	−128.96	15.1	38
32	47.18	−83.82	325	>85N	4.6	4	4	0	227.8	−26.7	228.1	−27.1	−38.43	−150.78	8.6	116
34	47.07	−83.73	315	>85S	20	7	3	1	195.6	−10.3	195.8	−10.6	−46.08	−106.77	14.7	52
35	47.07	−83.73	315	—	78	3	2	1	194.1	−11.4	194.3	−11.6	−46.99	−104.83	10.5	236
36	47.03	−83.69	320	—	14	6	2	0	209.9	−21.9	210.1	−22.2	−46.38	−129.10	27.9	82
37	47.03	−83.69	320	>85S	12	5	1	1	219.2	−1.0	219.5	−1.3	−32.33	−132.46	17.6	3064
38	47.02	−83.67	343	>85N	26	7	1	0	204.0	−1.0	204.2	−1.2	−39.00	−115.56	—	—
39	47.03	−83.61	330	—	27	5	2	0	190.0	−8.0	190.3	−8.1	−46.16	−96.48	31.0	67
41	47.02	−83.57	325	>85S	3.5	5	4	0	210.4	−24.5	210.7	−24.8	−47.41	−130.80	4.0	541
43	47.03	−83.55	325	90	1	5	4	0	205.4	−6.4	205.7	−6.8	−41.02	−118.59	2.5	1348
44	47.03	−83.55	310	85S	17	3	1	0	192.0	−5.0	192.3	−5.2	−44.30	−100.87	—	—
45	47.01	−83.51	341	85N	>22	9	5	0	200.6	−4.2	200.9	−4.4	−41.65	−112.05	12.8	37
46	47.02	−83.50	340	85N	17	6	1	3	206.0	−10.0	206.4	−10.3	−42.44	−120.31	21.1	76
48	46.99	−83.45	330	90	>31	6	1	1	203.8	−20.3	197.8	−16.4	−48.50	−110.64	49.8	385
49	46.86	−83.87	330	85N	11	7	2	0	202.3	−9.1	202.7	−9.1	−43.40	−115.37	4.3	3315
Normal polarity, unrotated																
16	47.27	−84.49	335	85S	2	7	3	1	31.9	22.9	31.7	23.0	45.84	48.03	16.7	40
22	47.33	−84.14	325	>85S	18	5	4	0	13.0	20.5	13.0	21.1	51.97	74.90	11.0	70
42	47.02	−83.57	320	>85S	14	7	2	1	24.7	8.9	25.0	9.2	42.41	61.59	6.5	614
Reversed polarity, rotated																
6	47.24	−84.65	311	55N	>52	7	5	0	209.4	18.7	208.8	18.7	−27.66	−117.08	16.8	22
7	47.23	−84.66	312	45N	22	14	4	1	210.0	6.9	209.5	6.9	−33.08	−120.51	16.3	27
8	47.21	−84.65	311	70N	28	10	4	0	205.3	12.9	204.8	12.9	−31.98	−114.04	6.9	178
10	47.20	−84.68	278	60N	>22	9	2	3	213.4	4.3	212.8	4.4	−32.81	−124.83	9.5	123
11	47.18	−84.68	289	70N	46	11	6	0	223.6	27.2	223.0	27.5	−17.25	−128.39	9.0	56
12	47.16	−84.70	340	55N	49	9	2	1	206.8	5.3	206.3	5.5	−35.02	−117.38	19.5	69
Normal polarity, rotated (?)																
4	47.24	−84.59	305	80N	31	8	7	0	35.5	15.2	35.1	15.1	40.45	46.89	6.0	103

TABLE A1 (*concluded*)

Site	Lat. (°)	Long. (°)	Trend (°)	Dip (°)	Width (m)	NT	NS	NR	D (°)	I (°)	D' (°)	I' (°)	Plat. (°)	Plong. (°)	α_{95} (°)	k
									PREISSAC NW DYKES							
									Reversed polarity							
28	47.09	−83.92	352	—	>5	6	5	0	112.2	−57.0	112.2	−56.9	−40.55	20.99	6.9	124
									Normal polarity							
27	47.03	−83.98	308	—	16	1	1	0	263.5	57.4	263.5	57.4	22.93	−142.20	—	—
31	47.08	−83.86	320	85S	15	6	3	0	290.0	56.5	290.0	56.4	38.78	−158.24	14.4	74
									SUDBURY DYKES							
25	47.18	−84.30	295	>85N	>25	6	6	0	266.7	16.3	266.4	16.6	3.81	−165.92	4.8	193
40	47.18	−83.59	300	—	60	5	5	0	259.2	14.9	259.5	14.3	−1.73	−160.93	5.8	176
									KEWEENAWAN DYKES							
1	47.39	−84.69	330	75N	15	7	7	0	299.9	54.2	299.3	54.8	44.13	−166.87	6.5	86
5	47.24	−84.62	318	70N	30	7	6	0	294.5	62.6	294.0	63.0	45.34	−152.73	6.7	100
24	47.11	−84.48	308	90	>11	7	6	0	296.2	39.7	295.9	40.1	33.92	−176.56	5.0	179
									DYKES OF UNKNOWN AFFILIATION							
9	47.21	−84.65	300	55N	28	7	2	3	268.8	−10.8	268.4	−10.0	−4.80	−176.87	33.2	12
13	47.16	−84.70	285	85N	3	3	3	0	279.8	38.2	279.2	38.9	22.02	−165.68	17.5	51
15	47.25	−84.55	335	75N	9	4	2	0	320.6	32.1	320.2	32.8	46.19	157.06	13.8	328
47	47.00	−83.48	335	80N	34	7	3	0	110.3	−0.6	110.7	0.1	−13.93	−8.96	9.6	165

NOTES: Dykes are assigned ages according to their paleomagnetic direction and petrographic characteristics. Lat. and Long. = latitude (N) and longitude (W) of the sampling site; Width = width of the dyke; NT = total number of samples demagnetized; NS = number of samples yielding stable end points; NR = number of additional samples yielding remagnetization circles only; D and I = declination and inclination of the remanence direction, and the corresponding primed values are those converted to a common site locality at long. 84°W and lat. 47°N; Plat. and Plong. = latitude (positive N) and longitude (positive E) of the paleopole derived from a site direction, and primed values in Table A2 refer to mean site values; α_{95} = radius of the 95% confidence circle about the mean sample/site direction; k = precision parameter. With respect to the site means, two of the Keweenawan dykes (sites 1 and 5) may have been rotated, thus giving an overestimate of the mean inclination value.

TABLE A2. Rejected site and site mean data

Site	Lat. (°)	Long. (°)	Trend (°)	Dip (°)	Width (m)	NT	ND	D (°)	I (°)	α_{95} (°)	k	Plat.′ (°)	Plong.′ (°)	D_p (°)	D_m (°)
			REJECTED SITES							SITE MEAN DATA					
										MATACHEWAN DYKES					
3	47.36	−84.66	315	80S	15	10				Unrotated reversed polarity					
14	47.24	−84.56	320	80N	8.5	4	17	204.1	−11.6	5.4	51.9	−43.961	−118.383	2.8	5.4
17	47.30	−84.39	350	85S	>14	5				Normal polarity					
18	47.32	−84.31	342	80S	26	4	3	23.2	17.9	17.9	48.6	45.79	57.17	6.8	13.2
										PREISSAC NW DYKES					
19	47.33	−84.25	325	85S	30	3				Normal polarity					
23	47.35	−84.11	330	85S	21	8	2	276.9	57.6	31.9	63.3	31.15	−149.66	34.3	46.8
26	47.17	−84.22	305	85N	>26	6				SUDBURY DYKES					
29	47.11	−83.90	295	90	5.5	4	2	262.9	15.5	15.4	265.3	0.983	−163.47	8.1	15.8
30	47.08	−83.86	325	85S	34	3				KEWEENAWAN DYKES					
33	47.09	−83.78	335	85N	18	6	3	296.5	52.7	18.0	48.1	40.96	−166.32	17.1	24.8

NOTES: See footnotes to Table A1. ND = number of sites; D_p and D_m are the semi-axes of the 95% ellipse of confidence about the paleopole.